HANYANG UNIVERSITY ERICA

Digital Logic Circuit Design

VHDL Assignment - 2

Student Information

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1. Purpose

The purpose of this assignment is to practice writing basic VHDL modules and testing their outputs in Logisim-evolution. I worked on the five problems from the VHDL_1 lecture material and saved screenshots of each circuit.

2. Problems

1. 4-Input AND and OR Gates
2. 8-to-1 Multiplexer
3. Showing Letters with a 7-Segment Display
4. 25-Person Voting Machine
5. Counting 1s and Displaying the Number

3. Test Method

For each problem, I placed the VHDL component in Logisim-evolution v3.9.0, connected input and output pins, changed the input values, and checked the output shown in the simulator. The next pages show the screenshots in problem order.

Note: The screenshots are example test cases, not a complete test of every possible input combination.

Problem 1. 4-Input AND and OR Gates

Task: Write VHDL code for a four-input AND output and a four-input OR output.

How it works

For the first circuit, there are four input signals (a, b, c, and d) and two output signals. The AND output is 1 only when all four inputs are 1. The OR output is 1 when at least one input is 1.

```
out1 <= a and b and c and d;
out2 <= a or b or c or d;
```

Simulation screenshot

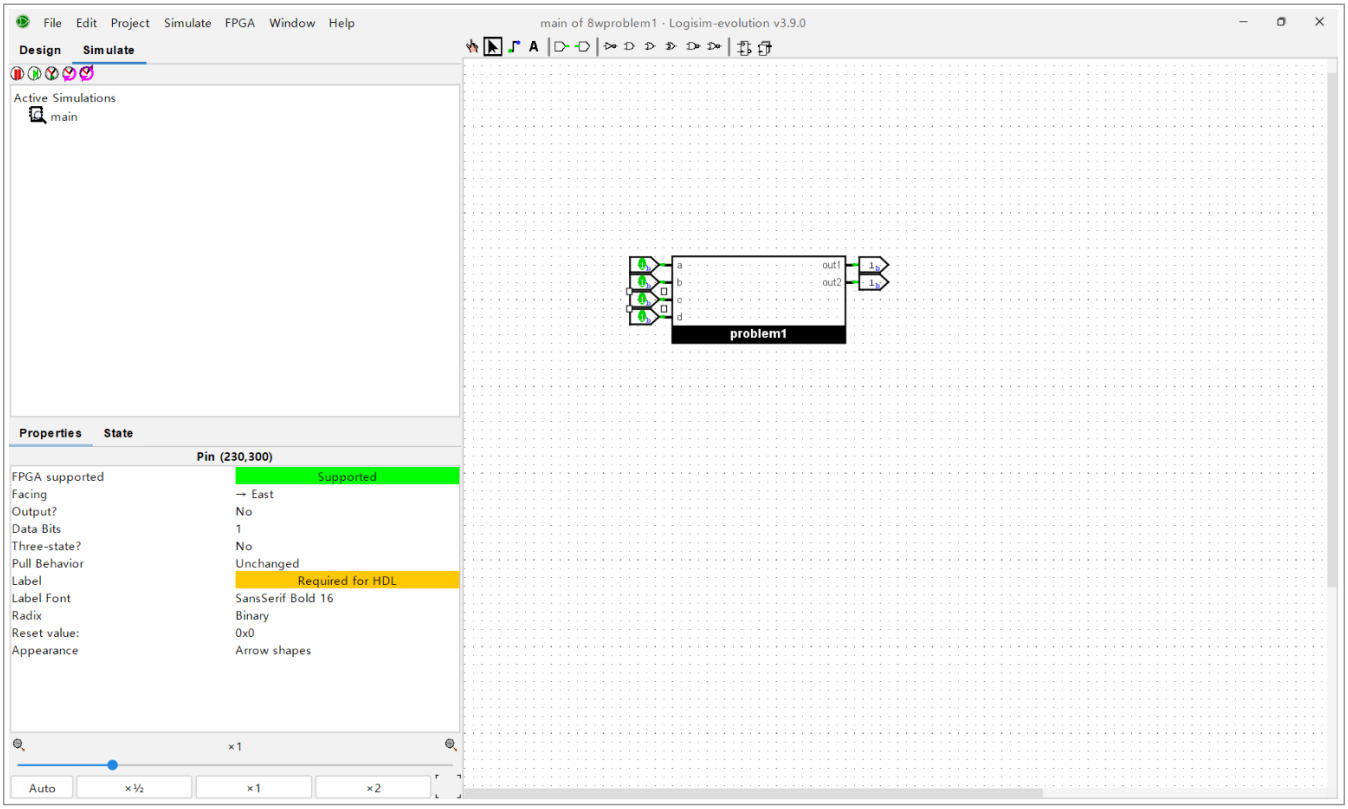


Figure 1. Logisim-evolution v3.9.0 test for Problem 1

Input condition	Expected output	Observed output
a = 1, b = 1, c = 1, d = 1	out1 = 1, out2 = 1	out1 = 1, out2 = 1

Result

I set all four inputs to 1 in Logisim-evolution. The two output pins both showed 1. This matches the AND and OR logic for this input condition.

Other cases: Only the 1111 input case is shown in the screenshot; the other input combinations are not included.

Problem 2. 8-to-1 Multiplexer

Task: Make an 8-to-1 multiplexer using VHDL.

How it works

An 8-to-1 multiplexer chooses one of eight data bits and sends it to a single output. A three-bit select signal is used, so the circuit can choose input 0 through input 7.

```
y = d(sel)    (concept of the selection)
```

Simulation screenshot

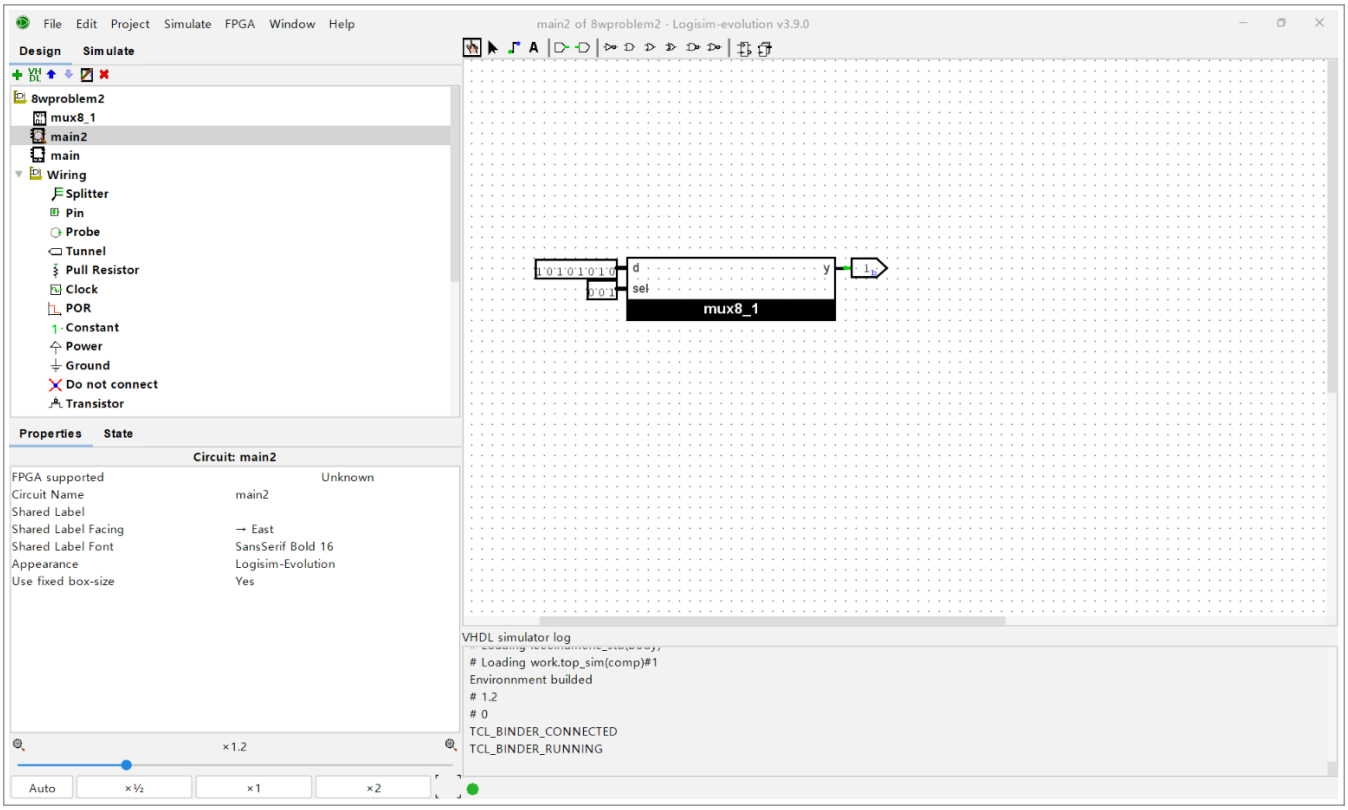


Figure 2. Logisim-evolution v3.9.0 test for Problem 2

Input condition	Expected output	Observed output
d = 10101010, sel = 001	y = d(1) = 1	y = 1

Result

I used the input pattern 10101010 and selected channel 1 (001). The output displayed 1, which is the value of the selected data bit.

Other cases: This picture shows one select value. It does not show the results for all eight channels.

Problem 3. Showing Letters with a 7-Segment Display

Task: Use a 7-segment display to show the letters a, b, c, d, f and g.

How it works

A 7-segment display is made of seven LED segments. Different combinations of segments are turned on to form numbers or letters. Here, the data input decides which letter pattern is sent to the display.

```
en = 1 : display enabled
dat : selects a character pattern
```

Simulation screenshot

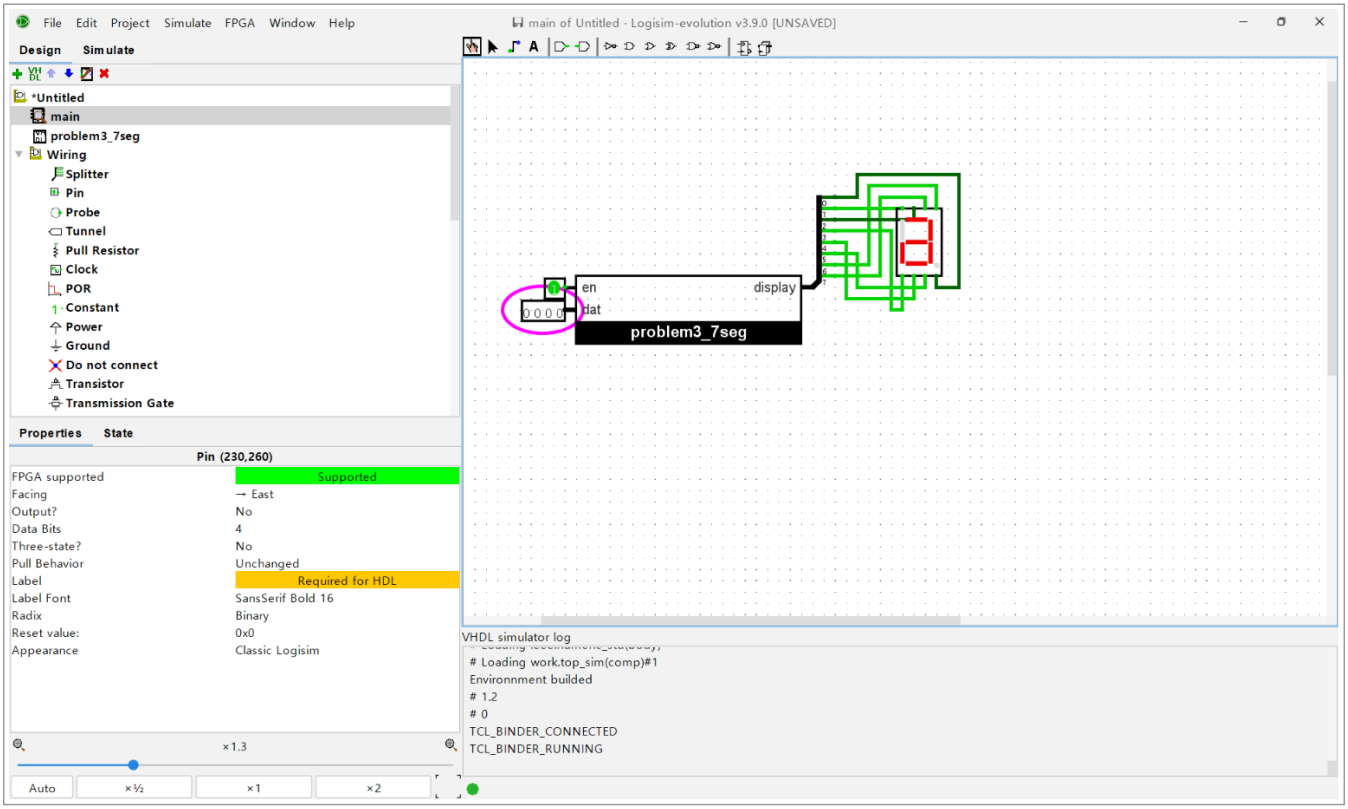


Figure 3. Logisim-evolution v3.9.0 test for Problem 3

Input condition	Expected output	Observed output
en = 1, dat = 0000	Letter a	Letter a

Result

When the enable signal was 1 and the data value was 0000, the display showed the shape of the letter a. This means the selected letter pattern was displayed correctly for the case shown.

Other cases: The screenshot does not show the other requested letters b, c, d, f or g.

Problem 4. 25-Person Voting Machine

Task: Design a voting machine for 25 people. The result is 1 when more than half vote 1.

How it works

Each of the 25 people has a one-bit input. The machine counts how many inputs are equal to 1. Since there are 25 voters, the result becomes 1 if at least 13 people vote 1.

```
result = 1 when number of 1s >= 13
result = 0 when number of 1s <= 12
```

Simulation screenshot

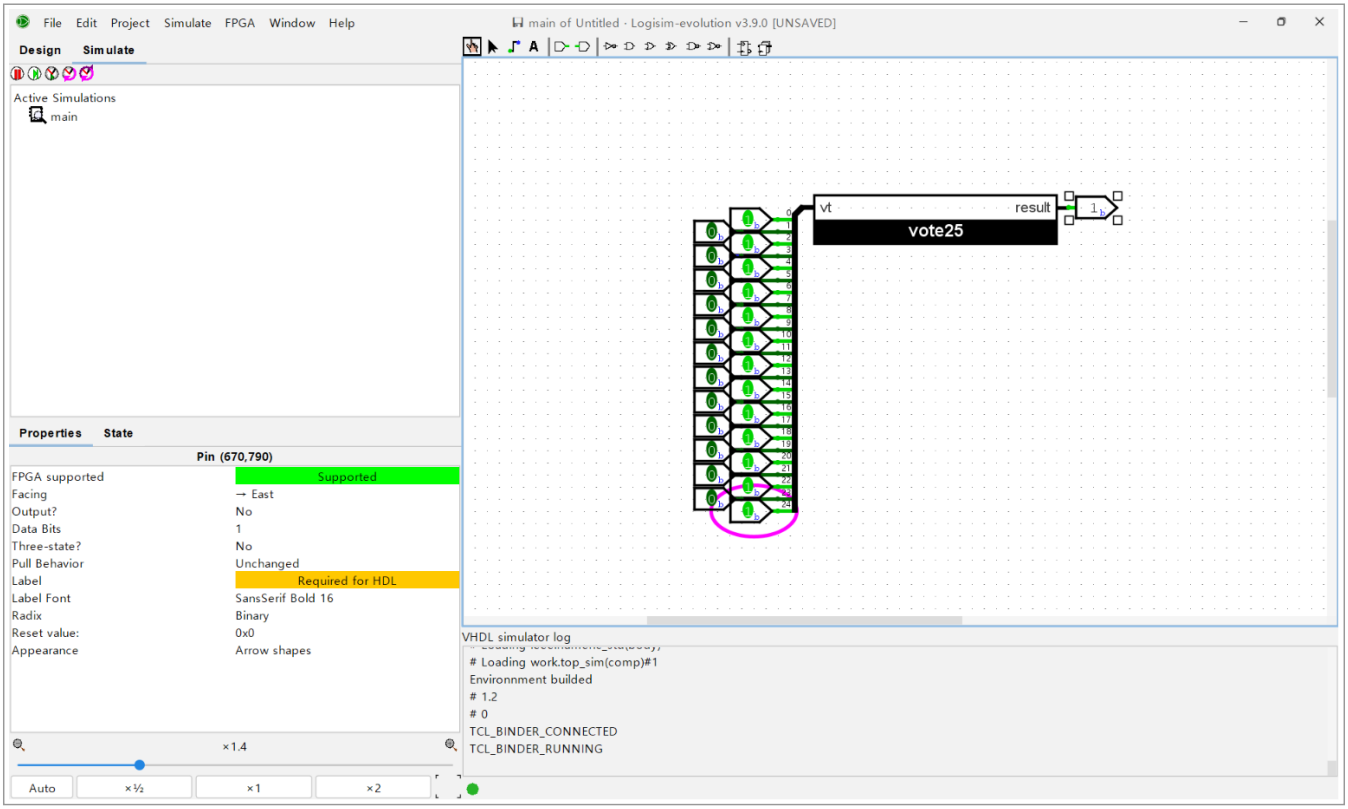


Figure 4. Logisim-evolution v3.9.0 test for Problem 4

Input condition	Expected output	Observed output
13 votes = 1, 12 votes = 0	result = 1	result = 1

Result

I tested the majority boundary with 13 inputs set to 1 and 12 inputs set to 0. The result pin displayed 1, so the circuit gave a pass result for more than half of the votes.

Other cases: A separate example with 12 or fewer votes set to 1 is not shown.

Problem 5. Counting 1s and Displaying the Number

Task: Count the 1 bits in a 13-bit input vt and show the result on a 7-segment display. When en is 0, turn the display off.

How it works

The circuit counts the number of 1 bits in vt and changes that count into a 7-segment pattern. The assignment asks for digits 0 to 9 and letters A to F for counts 10 to 15. The enable input controls whether the display is on.

```
count = total number of 1 bits in vt
en = 1 : show count    /    en = 0 : display off
```

Simulation screenshot

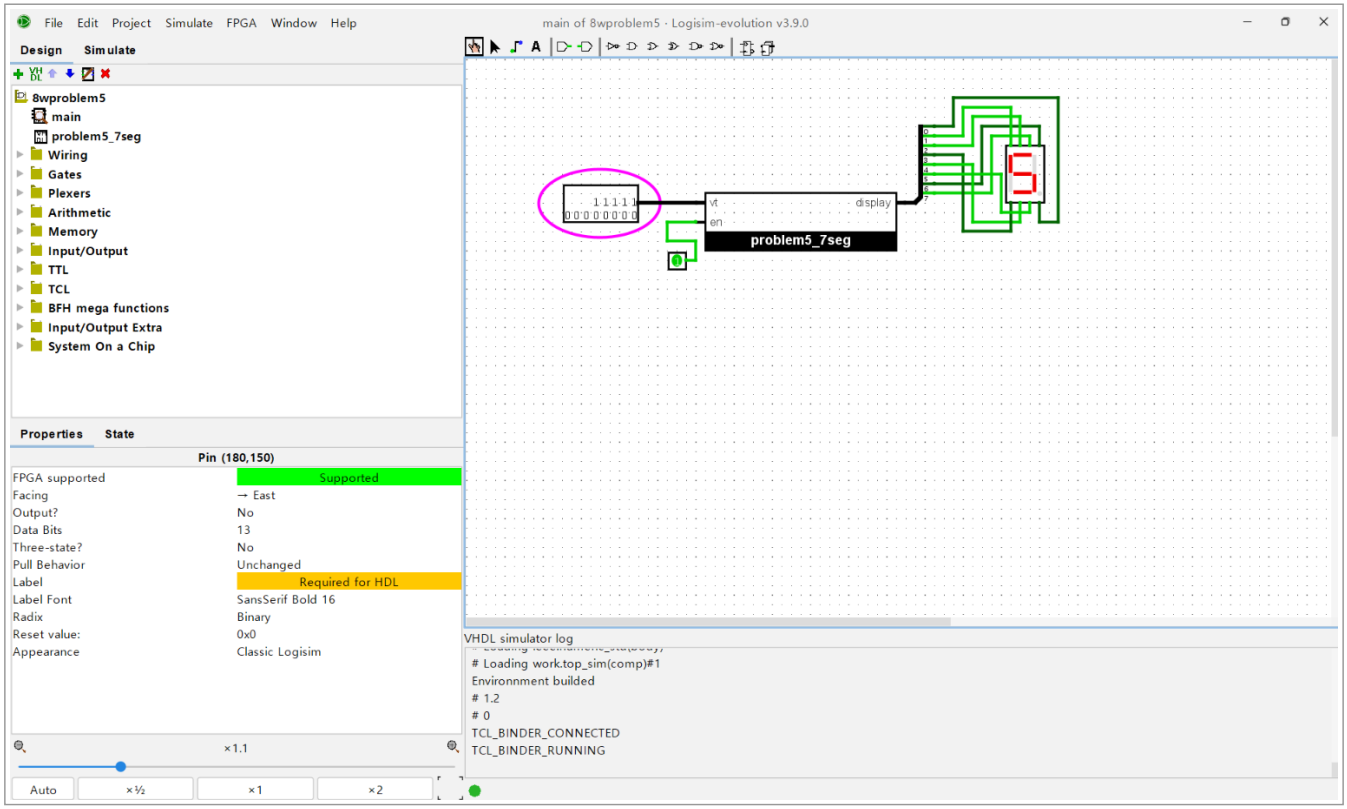


Figure 5. Logisim-evolution v3.9.0 test for Problem 5

Input condition	Expected output	Observed output
vt = 1111100000000, en = 1	Display 5	Display 5

Result

I entered a 13-bit value containing five 1s and set the enable signal to 1. The 7-segment display showed 5, which agrees with the number of 1 bits in the input.

Other cases: With only 13 input bits, the count can be 0 to 13. Therefore, counts 14 and 15 (E and F) cannot be reached with this input width. The en = 0 case is not pictured.